Roll No.: ..................................

## END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B. Tech. CSE) Semester: III

Name of the Course: Introduction to Cryptography Course Code: TCS-392

**Time: 3 Hours** **Maximum Marks: 100**

**Note:**

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

| | | |
|---|---|---|
| **Q1** | (20Marks) | |
| (a) | Explain in detail the OSI Security Architecture. How does it define the relationship between security attacks, services, and mechanisms? | CO1 |
| (b) | Encrypt the plaintext "SHE WENT TO THE STORE", using play fair cipher. The keyword to be used for creating diagraph for play fair cipher is "PROBLEMS".<br>Using hill cipher, encrypt message "ACT" using key= GYBNQKUR. | CO1 |
| (c) | Using Vigen'ere cipher, Encode the secret message by shifting the<br>–1st, 4th, 7th letters by 2<br>–The 2nd, 5th, 8th letters we shift by 9<br>–And the 3rd, 6th, 9th letters we shift by 21.<br>Encode: "GENERATED"<br>Given a simple substitution cipher where each letter is shifted by 3 positions (Caesar cipher), decrypt the following message: "FRQIOLFWLRQ". | CO1 |
| **Q2** | | |
| (a) | What is double DES? What kind of attack on double DES makes it useless?<br>Distinguish between diffusion and confusion. Explain Avalanche effect. | CO2 |
| (b) | Explain, with the help of neat and clean diagram, the working of a single round of DES with key generation. | CO2 |
| (c) | In Symmetric Key Cryptography, How a KDC can create a session key $K_{AB}$ between Alice and Bob (Simple protocol). | CO3 |
| **Q3** | | |
| (a) | Explain the Euler's Totient Function (φ(n) in detail. Derive the formula for computing φ(n) for both prime and composite numbers. Find the Euler's totient function (φ) of φ(21) and φ(35). | CO3 |

| | | |
|---|---|---|
| (b) | (i) Provide the detach of generating random numbers using Linear Congruential Method.<br>(ii) Generates the sequence using Linear Congruential Method given m=19, a=8, c=0, $X_0$=1. What is the period of this series? What should be ideal period of this series. | CO3 |
| (c) | State the facts of Euclidean algorithm. Find the greatest common divisor of (3486, 10292) and (2740, 1760) using Euclidean algorithm.<br>Find the multiplicative inverse of (3 mod 5) and (11 mod 26) using extended Euclidean algorithm. | CO4 |
| **Q4** | | |
| (a) | Explain the Key generation, Encryption and decryption steps of RSA. In RSA, Given p = 19, q = 23, and e = 3, find n, φ(n), and d. | CO4 |
| (b) | Consider that user A wants to communicate securely with user B using a symmetric key encryption algorithm. Describe in detail all the steps involved in the symmetric key exchange process. For each step, clearly specify:<br>• Which entity (A, B, or KDC) is sending the message<br>• Which entity is receiving the message<br>• The exact content of the message exchanged (in diagram form)<br>• The purpose of each step in establishing a shared symmetric key. | CO4 |
| (c) | What is message authentication? Explain the four possible ways in which a hash code is used to provide message authentication.<br>What is the number of padding bits if the length of the original message is 2590 bits? | CO5 |
| **Q5** | | |
| (a) | Explain the architecture of distributed Intrusion Detection System (IDS). Compare signature-based and anomaly-based detection methods with real-world examples. | CO5 |
| (b) | Briefly Explain the Intellectual Property Rights (IPR) in cyberspace. Discuss how copyrights, patents, and trademarks protect digital content and innovation. | CO6 |
| (c) | Quantitatively analyze cybercrime growth where reported cases increase by 25% per year from a base of 10,000 in 2020. Estimate case numbers for 2025. | CO6 |